

XRPL Validator — Daily Engineering Report • 2026-03-23

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **17 commits** (xrpl-validator 13, xrpl-sdk 4)

Engineering changes

- **feat: validation signing, manifest, peer network page, dashboard redesign** (1fad47c , xrpl-validator)
Validation Signing: - Sign validations with real ledger hashes from StatusChange ACCEPTED - Persistent Secp256k1 validator key at /mnt/xrpl-data/validator_seed.hex - Node ID: n9JqfkBDaNkid4t6uiKnJ7bLcTsQNgx5UyYkFeGpFQk6p8zCyJEc - Dedicated relay connection to s1.ripple.com for validation delivery - Fixed frame length bug (+2 was corrupting TLS stream) - Manifest module built (needs format debugging for propagation) ...
- **feat: RocksDB engine with 100% hit rate, auto-fetch missing accounts** (2369ef5 , xrpl-validator) Engine: - Switched from sled to RocksDB — instant startup, 55MB RAM, 2% CPU (sled pegged 100% CPU and 10GB RAM with 18.7M entries) - One-time migration tool: migrate_db binary (sled → RocksDB) - Direct binary balance extraction from raw XRPL serialized data (bypasses full decoder, avoids unknown field errors) - Background auto-fetch: missing accounts fetched from RPC and cached in RocksDB automatically — hit rate ...
- **fix: revert strict peer sig verification (failing for all peers)** (d45bcb4 , xrpl-validator) The session signature verification fails for every peer — likely a shared_value computation or key extraction bug. Reverted to warn-and-continue until the root cause is diagnosed.
- **fix: peer session signature verification now enforced** (2fb6ce8 , xrpl-validator) Root cause: peer Public-Key header is base58-encoded (n9K...) but we were hex-decoding it, getting garbage bytes, verification always failed.
- **fix: self-healing cache, 4x fetcher threads, balance extraction** (77c1861 , xrpl-validator)
- Bad cache entries auto-deleted and re-fetched from RPC - 4 concurrent fetcher threads (was 1) - Fetch dedup check before queuing - Balance extraction starts at offset 7 (skip LedgerEntryType value) - Accept zero-balance accounts (bit 62 check removed)
- **fix: dual validation relay (s1 + s2) for higher agreement score** (ce5bf5d , xrpl-validator) Validation sender now connects to both s1.ripple.com and s2.ripple.com simultaneously. When one drops, the other keeps delivering. Should reduce missed validations from ~8% to near 0%.
- **feat: add 4 remaining features to match rippled** (e872eba , xrpl-validator)
- Transaction relay: forward deduplicated TMTransaction to all relay peers - Serve ledger data: respond to TMGetLedger/TMGetObjectByHash from RocksDB - Amendment voting: include supported amendments in validation on flag ledgers - State hash computation: build SHAMap from RocksDB, compare to network account_hash

- **feat: full transaction execution, independent ledger close, RPCA consensus** (d7fe7ae , xrpl-validator)
- Transaction execution writes state back to RocksDB (fee deduction + XRP payments) - set_xrp_balance: in-place binary modification of AccountRoot balance field - extract_destination/extract_tx_amount: parse raw tx for Payment execution - Independent ledger hash computation (LWR\0 prefix, full header fields) - Transaction tree hash via SHAMap - RPCA consensus engine: mempool, Open → Establish → Accept state machine - ...
- **fix: disable state write-back to prevent hit rate drift** (f23c96b , xrpl-validator) Partial execution (fees + XRP payments only) was modifying cached network state in RocksDB, causing cumulative balance mismatches. Track execution in-memory only until full tx execution is complete.
- **feat: full ledger close with 33-type transaction execution pipeline** (e32c7d3 , xrpl-validator) Bridge between live engine (RocksDB/binary) and xrpl-ledger apply pipeline (LedgerState/JSON/SHAMap): - decode_raw_tx: binary tx → TxFields via xrpl-core codec - decode_ledger_object: binary ledger object → JSON - close_ledger: loads affected accounts from RocksDB, decodes to JSON, runs full apply_transaction_set (preflight → preclaim → apply → do_apply), computes account_hash + tx_hash + ledger_hash independently

Docs and operations

- **security: remove all hardcoded seeds from test code** (1d7c647 , xrpl-sdk)
- **security: remove all remaining seeds, fix zero-key fallbacks** (a405440 , xrpl-sdk)
- **security: audit fixes — enforce peer auth, gitignore hardening** (5c8beab , xrpl-validator)
- **security: zeroize secp256k1 account_private_bytes** (ff0df93 , xrpl-sdk)
- **security: bind localhost, env var paths, restrict visibility** (19d48ff , xrpl-validator)
- **security: Trail of Bits audit — fix all critical, high, medium issues** (2acef68 , xrpl-validator)
- **security: Trail of Bits audit — zeroize, constant-time, CI fixes** (6fa83d0 , xrpl-sdk)

Appendix — commits

Time	Repo	Commit	Subject
00:47	xrpl-validator	1fad47c	feat: validation signing, manifest, peer network page, dashboard redesign
05:57	xrpl-validator	2369ef5	feat: RocksDB engine with 100% hit rate, auto-fetch missing accounts
16:43	xrpl-sdk	1d7c647	security: remove all hardcoded seeds from test code
16:56	xrpl-sdk	a405440	security: remove all remaining seeds, fix zero-key fallbacks
16:56	xrpl-validator	5c8beab	security: audit fixes — enforce peer auth, gitignore hardening
17:02	xrpl-sdk	ff0df93	security: zeroize secp256k1 account_private_bytes
17:03	xrpl-validator	19d48ff	security: bind localhost, env var paths, restrict visibility
17:05	xrpl-validator	d45bcb4	fix: revert strict peer sig verification (failing for all peers)
21:54	xrpl-validator	2acef68	security: Trail of Bits audit — fix all critical, high, medium issues
21:54	xrpl-sdk	6fa83d0	security: Trail of Bits audit — zeroize, constant-time, CI fixes
21:59	xrpl-validator	2fb6ce8	fix: peer session signature verification now enforced
22:13	xrpl-validator	77c1861	fix: self-healing cache, 4x fetcher threads, balance extraction
22:21	xrpl-validator	ce5bf5d	fix: dual validation relay (s1 + s2) for higher agreement score
23:35	xrpl-validator	e872eba	feat: add 4 remaining features to match rippled
23:47	xrpl-validator	d7fe7ae	feat: full transaction execution, independent ledger close, RPCA consensus

Time	Repo	Commit	Subject
23:49	xrpl-validator	f23c96b	fix: disable state write-back to prevent hit rate drift
23:57	xrpl-validator	e32c7d3	feat: full ledger close with 33-type transaction execution pipeline